Study guide

- Read Saracino, §11.
- Know the definition of a *normal subgroup*, and the notation $N \triangleleft G$.
- Why are the following statements all equivalent to $H \triangleleft G$? (1) for all $g \in G, h \in H$, $ghg^{-1} \in H$; (2) $gHg^{-1} = H$ for all $g \in G$; (3) left cosets and right cosets are the same: $gH = Hg$ (Theorem 11.1).
- **Extrememly important definition:** The *quotient group* G/H, which is well-defined when $H \triangleleft G$ (Theorem 11.6).

1. Recall that $\mathrm{SL}(2,\mathbb{R})$ denotes the subgroup of $\mathrm{GL}(2,\mathbb{R})$ consisting of all 2×2 invertible matrices with determinant 1. Prove that $\mathrm{SL}(2,\mathbb{R})$ is normal in $\mathrm{GL}(2,\mathbb{R})$.

2. Prove that the intersection of any two normal subgroups in a group G is also a normal subgroup.

3. Suppose that G is a group, and N, H are subgroups with $N \trianglelefteq G$. Prove that the set

$$NH = \{nh : \ n \in N, h \in H\}$$

is a subgroup of G.

Note in both solutions above, I use the fact that H and N are closed under multiplication and inverse implicitly in several places. It is fine to leave these statements out of your solutions at this point, but it is still a good exercise to be aware of where you use them. See if you can spot the places in the solution above where these "closure" properties are used implicitly.

4. Suppose that G is an abelian group, and H is a subgroup. Prove that G/H is abelian.

5. Suppose that G is a group of order 21, and $g, h \in G$ are two elements of G such that $o(g) = 7$, $o(h) = 3$, and $hg = g^2h$. It follows from these assumptions that any element of G can be written uniquely as $g^i h^j$ for some $i \in \mathbb{Z}_7$ and $j \in \mathbb{Z}_3$ (you may assume this without proof, but you may find it interesting to think about why; ask me if you are interested).

   (a) Compute each of the following elements. Express each answer as $g^i h^j$, where $i \in \mathbb{Z}_7$ and $j \in \mathbb{Z}_3$.

   $$(gh)^2, \ (g^4h^2)(g^3h), \ (g^3h)^{-1}$$

   (b) Prove that $\langle g \rangle$ is a normal subgroup of G, but that $\langle h \rangle$ is not.

   (c) Show that $G/\langle g \rangle$ is a cyclic group of order 3.

   Comment: this description of G is very similar to our description of the dihedral groups D_n. Indeed, both are examples of a construction called a *semi-direct product*.

6. Let G be a group. Given any two elements $a, b \in G$, the *commutator* of a and b, denoted $[a, b]$ is defined to be

$$[a, b] = aba^{-1}b^{-1}.$$

   (a) Prove that G is abelian if and only if for all $a, b \in G$, $[a, b] = e$.

(b) Prove that for a normal subgroup $H \trianglelefteq G$, the quotient group G/H is abelian if and only if $[a, b] \in H$ for all $a, b \in G$.

7. Consider the quotient group $C = (\mathbb{R}, +)/(\mathbb{Z}, +)$.

   (a) Prove that for every integer n, there exists an order-n subgroup of C.

   (b) Prove that if $x \in \mathbb{R}$ is irrational, then the element $\mathbb{Z} + x$ of C has order ∞.

   (c) (Bonus; for extra credit) Prove that any finite subgroup of C is cyclic.

   Comment: the notation $\mathbb{Z} + x$ gets a bit clunky to write over and over again. You may prefer to instead use the "overline" notation $\overline{x}$ (as used in Chapter 9) for the coset of x.

8. Let G be a group, and $Z(G)$ denote the center of G (the set of all elements that commute with every element of G). Prove that if $G/Z(G)$ is a cyclic group, then G is abelian.

9. Suppose that G is a group, H is a normal subgroup of G with $|H| = 7$ and $[G : H] = 20$. Lagrange's theorem implies that if $x \in H$, then $x^7 = e$. Prove the converse: if $x \in G$ satisfies $x^7 = e$, then $x \in H$.

   Hint: use the quotient group.